



RecycleBot: Machine Learning Based Automated Waste Detection and Classification



Recycling Statistics

- Only 6% of 40 million tons of plastics produced is recycled in US.
- In US, between 2019 and 2020, plastics recycling decreased by 5.7%.
- 85% of plastic packaging goes to landfill.
- Globally, 400 million tons of plastic waste is produced annually.

Everyday millions of recyclable items are thrown in the trash because of lack of clarity if they are recyclable or not.



Design an easy-to-use web application that can be run on any device to guide the user if a specific item should be thrown in the recycle bin instead of putting in the trash.

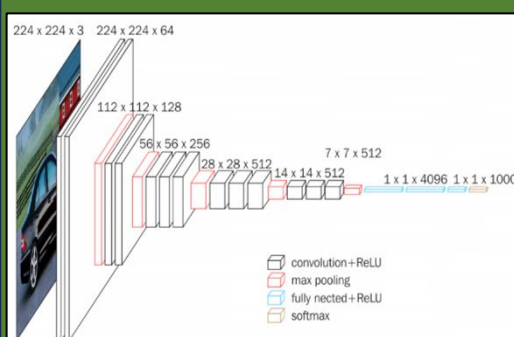
Challenge



Innovation

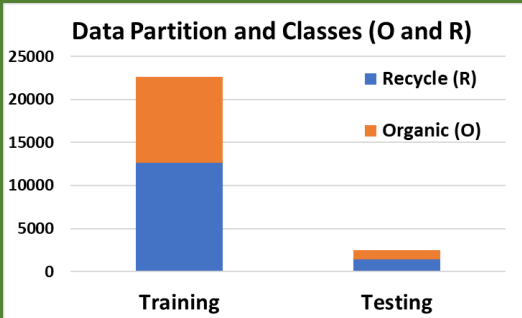
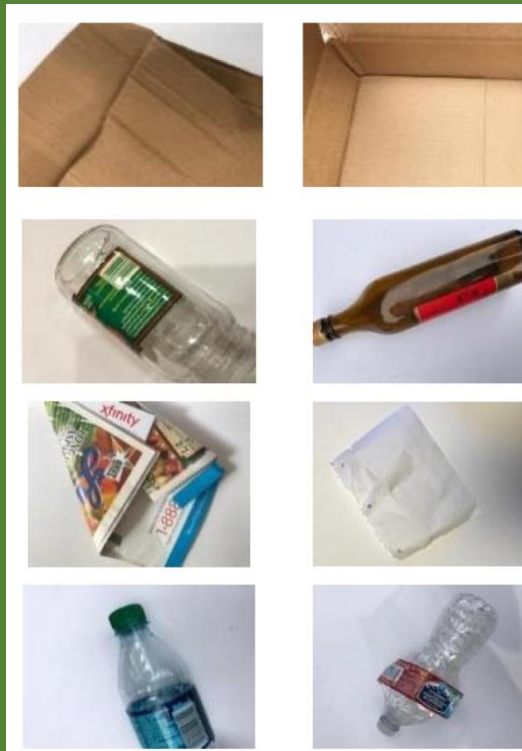
Machine Learning based image classification program is designed to identify if something is recyclable or organic.

Convolutional Neural Network (CNN) model built on VGG16 is trained and implemented in a web application, RecycleBot.



Data Libraries

- ImageNet Data set:
14,197,122 annotated images for transfer learning
- Data set – Waste Classification data (Kaggle):
containing 22,500 images of organic/recyclable objects.



Modeling Platform

- Python: Jupyter notebook
- GPU Computing
- Libraries – Pandas, Keras, Matplotlib, Scikit learn

Materials

- Camera
- Field Images

Research Study

How effective are the CNN models in detecting & classifying trash items? Identify the best model with optimal settings.

Independent Variables:

- Learning Rate (LR)
- Dropout Fraction (DO)
- Optimization Method (OPT)
- Epochs (training iterations)

Dependent Variables:

- Model Accuracy
- Classification error
 - Precision & Recall
 - F1 Score

Controlled Variables:

Resolution & size of the images

Control (Test Data set):

Pictures not used in the training.

Methodology

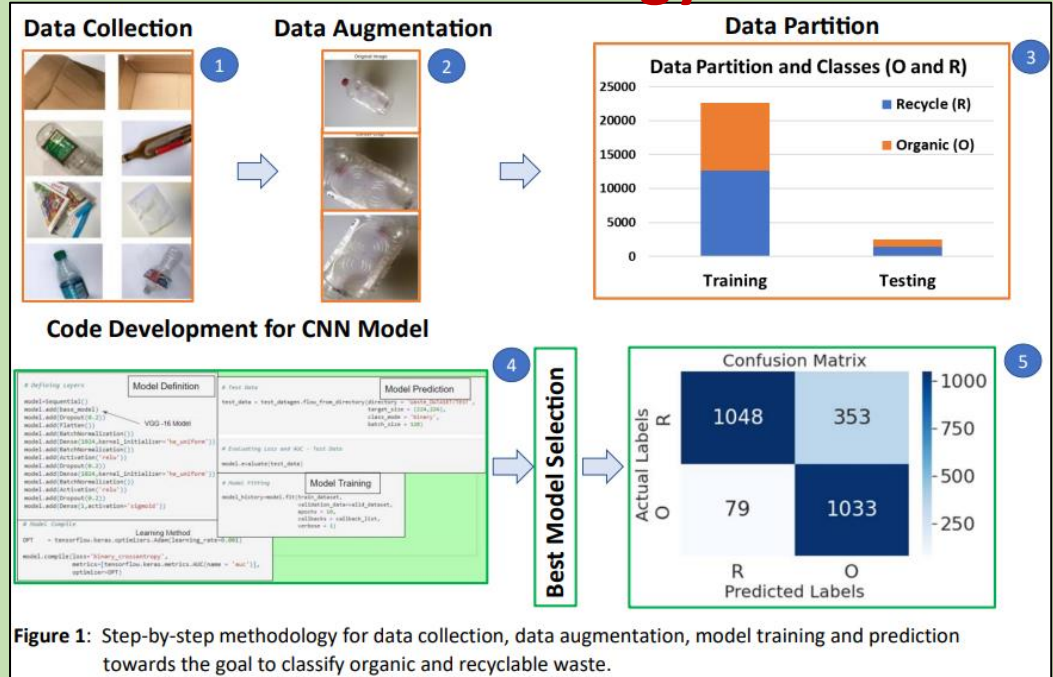


Figure 1: Step-by-step methodology for data collection, data augmentation, model training and prediction towards the goal to classify organic and recyclable waste.

Analysis and Results

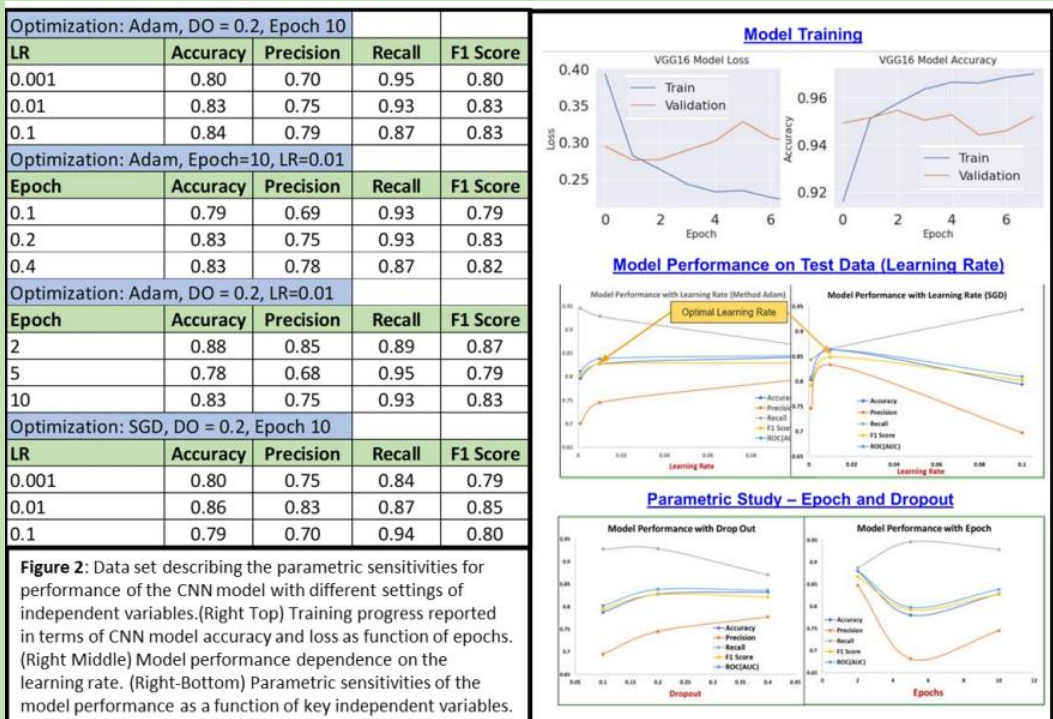


Figure 2: Data set describing the parametric sensitivities for performance of the CNN model with different settings of independent variables. (Right Top) Training progress reported in terms of CNN model accuracy and loss as function of epochs. (Right Middle) Model performance dependence on the learning rate. (Right-Bottom) Parametric sensitivities of the model performance as a function of key independent variables.

Implementation

Model Definition

Defining Layers

```
model=Sequential()
model.add(base_model)
model.add(Dropout(0.2))
model.add(Flatten())
model.add(BatchNormalization())
model.add(Dense(1024,kernel_initializer='he_uniform'))
model.add(BatchNormalization())
model.add(Activation('relu'))
model.add(Dropout(0.2))
model.add(Dense(1024,kernel_initializer='he_uniform'))
model.add(BatchNormalization())
model.add(Activation('relu'))
model.add(Dropout(0.2))
model.add(Dense(1,activation='sigmoid'))
```



Model Compile

```
T = tensorflow.keras.optimizers.Adam(learning_r
del.compile(loss='binary_crossentropy',
            metrics=[tensorflow.keras.metrics.AUC(
                optimizer=OPT)
```



Model Training

Model Fitting

```
model_history=model.fit(train_dataset,
                        validation_data=valid_dataset,
                        epochs = 10,
                        callbacks = callback_list,
                        verbose = 1)
```



Model Prediction

Test Data

```
test_data = test_datagen.flow_from_directory(directory = 'Waste_DAT
target_size = (224,224)
class_mode = 'binary',
batch_size = 128)
```

Evaluating Loss and AUC - Test Data

```
model.evaluate(test_data)
```

References

1. Mahanta, Jahnavi. "Introduction to Neural Networks, Advantaged and Applications.", 10 July 2017.
2. Sha, Pratik. "Difference between Python and Java.", 31 Mar. 2020.
3. Ching, Collin. "How to build and image classifier for waste sorting, towards datascience, 27 Mar. 2019.
4. Siegel, Joshua, et al., editors. "RUCycle." RUCycle, 2017, devpost/software/rucycle.

Field Application

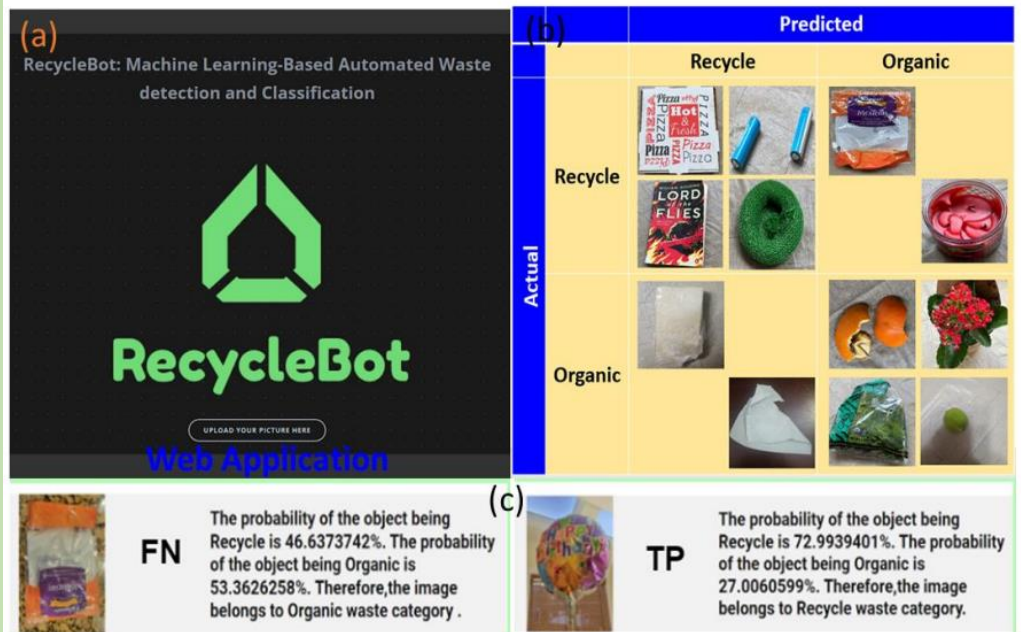


Figure 3: (a) RecycleBot: Web Application Interface (b) Confusion Matrix for classification of field data using RecycleBot (c) A representative output screen predicting the likelihood of an item to be organic or recyclable.

Conclusions

Image classification using Convolutional Neural Network (CNN) is effective in classifying trash into organic & recycle.

- VGG16 enhanced with three deep layers performs at > 90% accuracy in predicting the recycle objects for training data. On 2500 test images: 83% accurate (17% false positives/negatives).
- Learning rate is an important parameter in model training and an optimum value of 0.01 was used to arrive at the best predictive model (F1 score of 83%). Drop Out of 0.2 is optimal.
- The classification model is implemented as a web-app RecycleBot; successfully demonstrated on field images.

Environmental Applications

- **Waste Management:** RecycleBot can play a key role in driving ease & fraction of recycle content towards improving sustainability of the environment.
- **Biohazard Safety:** Model can be customized to identify and segregate medical and infectious waste.
- **Recycle Culture:** Facilitates awareness & culture of recycling!

Future Research

- Extend the model to include other categories of classification.
- **Composite Pictures:** Implement image segmentation to classify images with multiple objects for efficiency.
- **Smart Recycle Bin:** Integrate RecycleBot into the physical recycle bin to further enhance the automation of trash classification.